

JavaScript Events - Complete Notes with Examples

This document covers the most commonly used JavaScript events with simple explanations and practice examples.

What is an Event?

An event is an action performed by the user or browser such as clicking a button, typing in a textbox, moving the mouse, loading a page, or submitting a form. JavaScript listens for these actions and executes code.

Ways to Add Events

1. Inline:

```
<button onclick="show()">Click</button>
```

2. onclick property:

```
document.getElementById("btn").onclick=function(){  
    alert("Clicked");  
}
```

3. addEventListener() (Recommended):

```
document.getElementById("btn").addEventListener("click",function(  
) {  
    alert("Clicked");  
});
```

Common Events

Mouse Events:

click, dblclick, mouseover, mouseout, mousemove, mousedown, mouseup

Keyboard Events:

keydown, keyup

Form Events:

input, change, focus, blur, submit, reset

Window Events:

load, resize, scroll

Clipboard Events:

copy, cut, paste

Examples

Click

```
button.addEventListener("click",function(){  
    alert("Button Clicked");  
});
```

Double Click

```
button.addEventListener("dblclick",function(){  
    alert("Double Clicked");  
});
```

Mouse Over

```
box.addEventListener("mouseover",function(){  
    this.style.background="green";  
});
```

Mouse Out

```
box.addEventListener("mouseout",function(){  
    this.style.background="red";  
});
```

Mouse Move

```
document.addEventListener("mousemove",function(e){  
    console.log(e.clientX,e.clientY);  
});
```

Key Down

```
input.addEventListener("keydown",function(e){  
    console.log(e.key);  
});
```

Key Up

```
input.addEventListener("keyup",function(){  
    console.log(this.value);  
});
```

Input

```
input.addEventListener("input",function(){  
    console.log(this.value);  
});
```

Change

```
select.addEventListener("change",function(){
```

```
        alert(this.value);  
    });
```

Focus

```
input.addEventListener("focus",function(){  
    this.style.background="yellow";  
});
```

Blur

```
input.addEventListener("blur",function(){  
    this.style.background="white";  
});
```

Submit

```
form.addEventListener("submit",function(e){  
    e.preventDefault();  
    alert("Form Submitted");  
});
```

Reset

```
form.addEventListener("reset",function(){  
    alert("Form Reset");  
});
```

Load

```
window.addEventListener("load",function(){
```

```
    console.log("Page Loaded");  
  });
```

Resize

```
window.addEventListener("resize",function(){  
    console.log(window.innerWidth);  
});
```

Scroll

```
window.addEventListener("scroll",function(){  
    console.log(window.scrollY);  
});
```

Copy / Cut / Paste

```
document.addEventListener("copy",()=>alert("Copied"));  
document.addEventListener("cut",()=>alert("Cut"));  
document.addEventListener("paste",()=>alert("Pasted"));
```

Event Object

Useful properties:

event.target

event.type

event.key

event.clientX

event.clientY

preventDefault()

```
link.addEventListener("click",function(e){  
    e.preventDefault();
```

```
});
```

stopPropagation()

```
child.addEventListener("click",function(e){  
    e.stopPropagation();  
});
```

Event Bubbling vs Capturing

Bubbling: Child -> Parent (default)

Capturing: Parent -> Child (use true as third parameter)

```
parent.addEventListener("click",handler,true);
```

removeEventListener()

```
function hello(){  
    alert("Hello");  
}  
  
btn.addEventListener("click",hello);  
  
btn.removeEventListener("click",hello);
```

Mini Practice Projects

1. Counter App
2. Live Character Counter
3. Background Color Changer
4. Password Show/Hide
5. Keyboard Detector
6. Student Result System
7. Calculator
8. Todo List
9. Movie Ticket Booking
10. Traffic Light Simulator

Interview Questions

- What is an event?
- Difference between `onclick` and `addEventListener()`?
- What is the event object?
- What is `preventDefault()`?
- What is `stopPropagation()`?
- What is event bubbling?
- What is event capturing?
- Difference between input and change events?